MINE FLOOR MATERIAL RECOVERY SHOULD BECOME PART OF LIFE OF MINE PLAN

(Au & Ni Prices Updated on 13 08 2026)

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Ausvac Mining PL Mobile Supersucker - Krzysztof (Kris) Biegaj
Kalgoorlie Mining Expo 23 10 2003 - Western Australia







UG Nickel Mine in Western Australia – 2007 in Kambalda Area

- Suction hose Dia. 200mm/8 inch
- ~ 0.5 m^+ thick high-grade ore left behind on mine floors had been vacuumed on gold and nickel mines in Australia by Ausvac Mining PL in 2003-2011 *(Eight Hands-on Projects by Kris Biegaj)*
- When using conventional loaders/scrapers due to the "milling effect", presence of water and gravity force - **significant grade enhancement occurs in the ore left behind on mine floors**

INTRODUCTION

Billions of Dollars have been, and continue to be wasted by Companies and Governments Globally due to the lack of legislation for the recovery of already broken ore left behind/wasted on mine floors - never to be recovered

Recovery of broken ore left behind on mine floors will provide:

- Significant additional revenue for mining companies – refer to slides No **13 & 14** **showing graphs with extra monthly & yearly revenue on typical smaller/medium size Australian nickel & gold mines** with the Ausvac Mining PL vacuuming technology hands-on developed in 2003 – 2011 *(Excel Spreadsheets on Page No 14)*
  - Detailed calculations of Revenue Loss Recovery in Aus$ by Vacuuming of Mine Floors (LHS)
  - Vacuuming Costs Calculations (RHS)
- **Additional Royalties for Governments** – including **Compensation Funds for Indigenous Australians**

- Additional funds for rehabilitation of mined waste disposed off on surface
  - Including that portion of waste generated whilst accessing in waste & defining the portion of **broken ore left behind on mine floors**

House vacuuming invented and implemented globally **>120 years ago**

Mine floor material vacuuming machines and technology were first invented and used decades ago in South Africa and in only few other countries

The Ausvac Mining PL vacuuming machine is based on the late 1960-ties technology/machines designed to mine opals in Coober Pedy Region in South Australia

Highly-efficient truly mobile and modern mine floor material vacuuming machine developed in the Kalgoorlie Goldfields + *(Western Australia – seven (7) projects and one (1) project in Bendigo – Victoria)*, has been available since 2003+

- It can be equipped to be remotely-controlled operation in open stopes
- Whilst now still diesel-powered but easy to be converted to be electrically-powered
- When converted to be electrically powered - highly suitable to be operated in mines located in high-altitude
  - Including South America

Mining Project Stages

- Exploration, Pre-feasibility study, Feasibility study, Mineral Resources/Ore Reserves/Life of Mine Plans
- Approvals and finance secured/obtained
- Development in waste to access ore and disposal off mined waste, including diamond drill/sludge ore definition drilling cuddies *(excavated in waste)*
- Mining
- Rehabilitation of mined waste disposed off on surface and mine closure

Mining – What is being missed

- Cleaning - vacuuming/vamping of mine floors is globally not yet part of the mining cycle
- The mine reconciliation factors continue to be used by mining companies with inclusion of the high-grade broken ore left behind on mine floors
  - And permitted by Mining Legislations Globally
- Floors of the excavations to be cleaned/vamped/vacuumed
  - Ore drives
  - Stopes (**remotely controlled vacuuming only in open stopes**)
    - Footwall of low-angle dipping stopes
    - Top of last lift in cut & fill/under-hand cut & fill mining
    - Top of mine pillars left behind for geotechnical reasons
  - Underground sumps
- Selective mining of high-grade ore in open pits

- Highly efficient cleaning of mills & sumps on surface

Mine floor cleaning/vacuuming should become part of Mining Cycle

- After appropriate testwork, metals/precious stones loss to mine floors should be modelled and included in project studies to determine its impact

- Mine floors material recovery should be included in Pre-feasibility/Feasibility Studies and Ore Resources/Reserves Estimation/Life of Mine Plans
- **left behind on mine f**
- Timing - Mine floors material recovery should be planned & conducted whilst mine ventilation/fresh air and mine water/compressed air supply/dewatering and all infrastructures are fully operational
  - Not after mine is flooded/put on care and maintenance

Vacuuming Technology/Productivity Development by Ausvac Mining PL

- When started mine floor material recovery in 2003 no more than five (5) tonnes/shift from ore drives could be achieved with two operators
  - August/October 2003 Otter-Juan Nickle Mine WA – **Project # 2** & Bendigo Gold Mine, Victoria State - **Project #3**
  - ~ 0.5 m^{+} thick broken ore normally left behind on mine floors when using conventional LHD equipment
- After eight (8) hands-on projects & development of the mine floors material recovery system - now 40 t/shift are easily achievable with two (2) operators
  - With additional vacuuming capabilities available
  - Continuous roster on one or two shifts per day
- Detailed vacuuming costs calculations are on Page No 15
  - RHS Excel Spreadsheet

Summary of Gold & Nickel 8 (eight) mine floor vacuuming projects in Australia 2003 – 2011 by Ausvac Mining PL

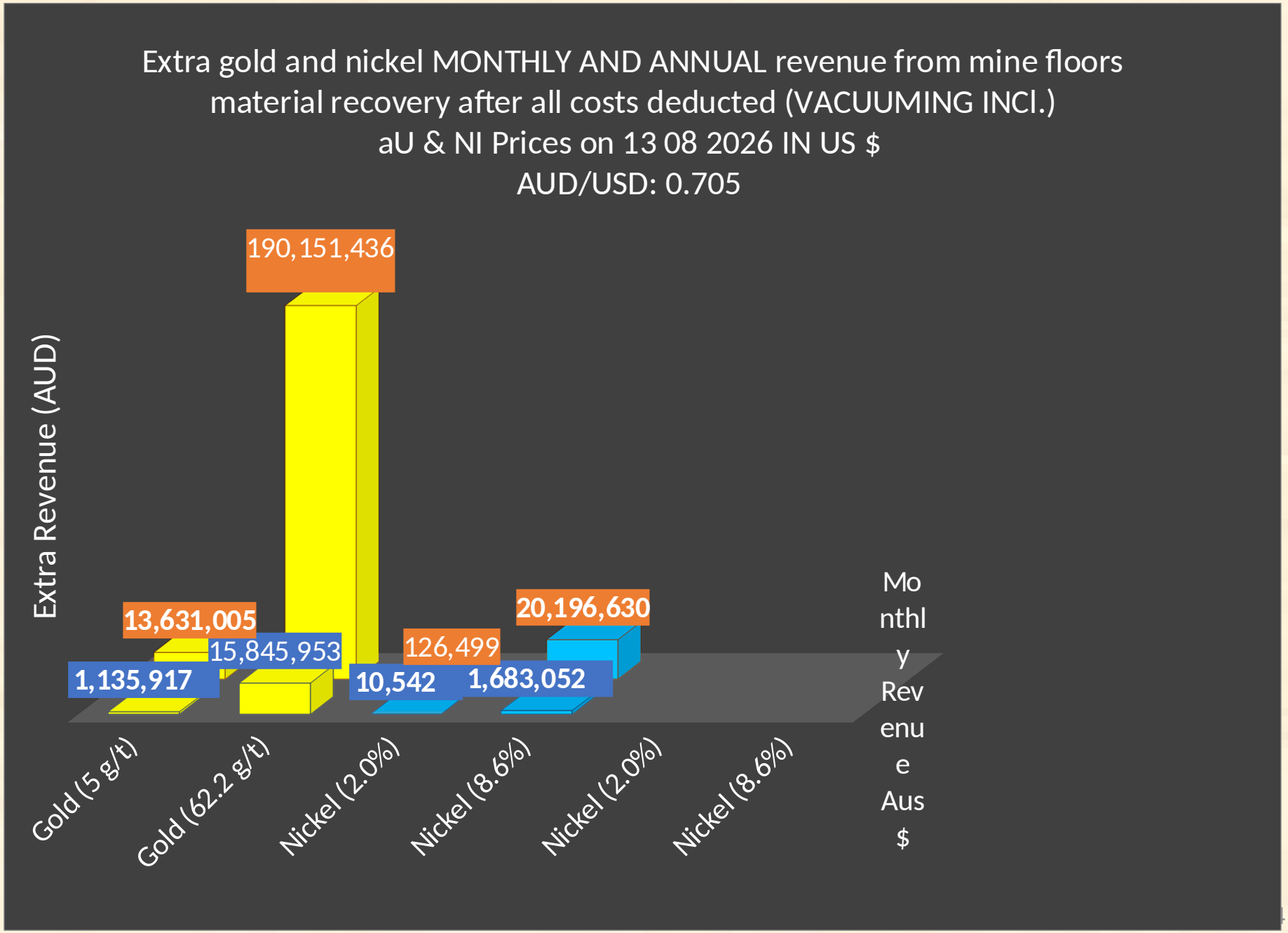
- Projects – *Ref. to Page 17 App. 12B Vacuuming Projects Map*
  - Mt Pleasant WA, May 2003 **– Gold #1** Project – vehicles clean-up sump on surface
  - Otter-Juan Mine UG WA, August 2003 – **Nickel #2** Project
  - Bendigo Mine UG Victoria, October 2003 – **Gold #3** Project
  - Cosmos Mine UG WA, December 2003 – **Nickel #4** Project
  - Agnew Mine UG WA, February 2004 – **Gold #5** Project
  - Long-Victor UG Complex WA, October 2005 - **Nickel #6** Project
  - Redross UG Widgiemooltha WA, February 2007 – **Nickel #7** Project
  - Harlequin Mine UG CNGC WA, November 2010 Plus - **Gold #8** Project
- Ore Grades Vacuumed
  - **Gold**: one (1) oz to two (2) oz +++; **Four (4)** Projects
  - **Nickel:** Average 8 % +; **Four (4)** Projects
- **left behind on mine f**

Mine Floor Sample in New In-Stope Decline

- Sample of Au grade of mine floor material on in-stope decline – CNGC Harlequin Gold Mine in Norseman WA
  - Grade: **40.02 g/t**
  - Total weight of sample: 131.9 kg
    - Last five (5) bottom layers samples across the decline's floor conducted with small vacuum cleaner to the solid floor
- **"Low"** grade due to **low milling factor** by conventional equipment & **low grade enhancement by** water and gravity force in **newly** excavated in-stope decline
  - **Details are on Page 17** – Sample conducted by Kris Biegaj *(then Mining Manager)* at the CNGC Harlequin Gold Mine in 2001

Extra Additional Revenue from Mine Floor Vacuuming

13 08 2026				
Metal	Grade	Monthly Revenue Aus$	Annual Revenue Aus$	Gold and Nickel prices on 13 08 2026 in US $/oz/mT
Gold	Gold (5 g/t)	1,135,917	13,631,005	4,387.50
Gold	Gold (62.2 g/t)	15,845,953	190,151,436	
Nickel	Nickel (2.0%)	10,542	126,499	16,849.6
Nickel	Nickel (8.6%)	1,683,052	20,196,630	


Extra gold and nickel MONTHLY AND ANNUAL revenue from mine floors
material recovery after all costs deducted (VACUUMING INCl.)
aU & NI Prices on 13 08 2026 IN US $
AUD/USD: 0.705
Extra Revenue (AUD)
190,151,436
13,631,005
15,845,953
126,499
20,196,630
1,135,917
10,542
1,683,052
Gold (5 g/t)
Gold (62.2 g/t)
Nickel (2.0%)
Nickel (8.6%)
Nickel (2.0%)
Nickel (8.6%)
Monthly Revenue Aus $

Extra Revenue from Mine Floors & Vacuuming Costs Calculation – in Excel Worksheets below

All Costs included i.e. Mine Floor Vacuuming, Ore Transportation, Milling & Roasting where applicable

Additional Extra Revenue from other metals (Copper +++) recovered by vacuuming **not included** in the Calculations

Typical length & width of mining excavations on Au & Ni mines during the (8) eight Ausvac Mining PL mine floor material vacuuming technology development Projects in Australia 2003 – 2011 used for calculations



Microsoft Excel Worksheet



Microsoft Excel Worksheet

EVALUATION OF BROKEN FLOOR STOCKS – OPERATING MINES

- Sampling and testwork should be undertaken in support of mine designs, Feasibility studies and LOM plans

- The Australasian Joint Ore Reserves Committee Code (JORC, 2012) must be applied in the context of Sections 37-41 and Table 1

ECONOMIC VALUE OF BROKEN ORE RECOVERY FROM MINE FLOORS - ORE RESOURCES/RESERVES ESTIMATION/LOM PLANS

- Modifications to the JORC Code are needed to address the challenge of potential revenue loss

- Where appropriate, vacuuming must be introduced to new and operating mines

Appendices



App 1 Kris Biegaj - Mining Expo Kalgoorlie Western Australia - 23 10 2003.JPG



App 2 High-grade Nickel Mine in Western Australia - Supersucker Towing - Jan 2004.JPG



App 3 - Mine Floor vacuuming - Nickel Mine near Kambalda - WA 2007.JPG



App 4-Discharging vacuumed floor material into Bobcat-Nickle Mine 2007.JPG



App 6 Surface Mining Equipement Sump near Kalgoorlie before clean-up 2003.JPG



App 6A - Mining Equipment Surface Clean-up Sump - before clean-up - 2003.JPG



App 7-Mining Vehicles Surface Clean-up Sump after clean-up 2003.JPG



App 8-Surface Sump on Gold Mine after Clean-up with Supersucker-2003.JPG



App 10 Other Applications.jpg



App 11 Winzing Plus Mining Aplications.jpg



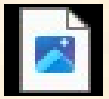
App11A - DD hole Bulen West 2001.jpg



App12 Floor Material Sample by Kris Biegaj-2001.jpg



App 12A Floor Material Sampling Details - 2001.jpg



App 12B - Ausvac Mining PL Projects Map in Australia jpg.jpg



Supersucker - Giant Mobile Vacuum Cleaner Constr. Details.jpg



Adobe Acrobat Document